

Divya Gupta

Software Engineer | Full Stack & AI Systems

 divya2207.work@gmail.com  Germany  linkedin.com/in/divya-gupta-3a505617a

EXPERIENCE

CISPA , Research Assistant

- Developed and maintained secure system components using **Haskell, Angular, D3.js, CSS, and TypeScript**, reducing system errors by **25%** and enhancing overall software robustness.
- Led the design and implementation of **advanced graph visualization tools**, increasing real-time data rendering speed by **40%** and enabling scalable exploration of complex datasets.
- Improved **UI/UX workflows** for research applications, reducing analysis time by **20%** and driving higher adoption rates of internal tools.
- Contributed code improvements, feature enhancements, and bug fixes to the open-source **Tamarin-Prover** project, strengthening capabilities for **secure protocol verification**.

08/2024 – Present
Saarbrücken, Germany

FICO, Software Engineer - I

- Designed and deployed **high-performance backend APIs** using **Java, Spring Boot, and SQL** for **Falcon**, a fraud detection system processing **millions of daily transactions**.
- Enhanced **fraud detection algorithms** and implemented **robust error-handling mechanisms**, reducing false positives by **15%** and improving transaction throughput reliability.
- Optimized **API performance**, reducing response latency by **~20%** and delivering a smoother end-user experience for **banking clients**.
- Achieved **90%+ unit test coverage** with **JUnit and Mockito**, improving system reliability and reducing production defects.
- Eliminated **95% of critical code vulnerabilities** by applying **SONAR-based static code analysis**, strengthening overall **application security**.
- Implemented robust **unit testing strategies** to enhance performance, scalability, and reliability of applications

06/2023 – 02/2024
Bangalore, India

FICO, Associate Software Engineer

- Spearheaded the **design and implementation of REST APIs, Websockets, SOAP APIs and OAUTH** using **Java, Bootstrap, Tailwind CSS** and **AngularJS(NGRX)** for **FAWB (FICO Analytical Workbench)**, enhancing scalability and maintainability of core modules.
- Refactored **API logic** and optimized **SQL queries**, reducing defect rates by **20%** and improving performance in high-volume data processing.
- Redesigned and **optimized UI/UX components** to streamline user workflows, decreasing average task completion time by **15%** and improving customer usability.
- Collaborated with **Agile-Scrum teams** throughout sprint cycles, achieving **100% on-time delivery** of prioritized features.
- Received **Special Recognition Award** for driving impactful improvements in **product architecture** and **delivery efficiency**.
- Contributed to **real-time data rendering**.

06/2022 – 06/2023
Bangalore, India

FICO, Software Engineering Intern

06/2021 – 07/2022
Bangalore, India

- Contributed to the **Origination Manager project** within **FAWB**, enabling faster onboarding of new features and reducing manual intervention in workflow setup.
- Resolved **125+ product defects** and delivered multiple **user stories per sprint**, reducing overall backlog by **15%** and improving product stability.
- Developed **responsive UI components** using **JavaScript, SCSS, and API integrations** with **Wavemaker**, enhancing platform functionality and user experience.
- Applied **AWS Cloud Practitioner knowledge** to optimize scalability, deployment processes, and system reliability.
- Conducted **knowledge transfer sessions** and collaborated with cross-functional teams, accelerating onboarding and improving delivery efficiency.

EDUCATION

Master of Science in Computer Science, Universität des Saarlandes

04/2024
Saarbrücken, Germany

LANGUAGES

English (IELTS - 8) ● ● ● ● ● German (B1) ● ● ● ● ●

PROJECTS

AI Interview Simulator (PrepBot)

- Developing a full-stack AI interview preparation platform with speech-to-text transcription and automated feedback generation (*Master Thesis – Ongoing*)
- Focused on human-computer interaction and personalized AI feedback systems

[GitHub: <https://github.com/Divya19gupta/PrepBot-InterviewSimulator>]
[MERN Stack, OpenAI, AssemblyAI]

Hindi Morphological Inflection: CRF vs LSTM comparision

- Implemented and evaluated CRF and LSTM models on Hindi morphological inflection using the UniMorph dataset.
- Completed as a seminar project with a comparative analysis of classical vs. deep learning approaches

[GitHub: <https://github.com/Divya19gupta/hindi-pluralization-model-comparision>]
[Python, CRF, LSTM, PyTorch]

SKILLS

Java, JavaScript, TypeScript, Python, LangChain, PyTorch, Hugging Face Transformers, SQL, JUnit, Mockito, Spring Boot, Spring Data, Spring Security, AngularJS, ReactJS, Next.js, Express.js, Node.js, Tailwind CSS, D3.js, Viz.js, Data Structure and Algorithms, REST APIs, Graph Visualization, HTML, SCSS, Bootstrap, CSS, JSON, WebSockets, OpenAI Whisper API, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Integration, Multimodal AI Applications, AWS (Cloud Practitioner), Docker, Jenkins, Git, Maven, CI/CD Pipelines, Cypress, SONAR, Unit & Integration Testing, Static Code Analysis, MongoDB, Agile-Scrum, Jira, Confluence, Cross-functional Team Collaboration, Figma.